

Jiacheng LIU

Postgraduate Student of Electronic and Computer Engineering

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Education

- 2022.09 – present **Postgraduate Student** in Electronic and Computer Engineering.
The Hong Kong University of Science and Technology, Hong Kong
- Advisor: Prof. Qiming SHAO
 - CGA: 3.984/4.3
- 2023.11 – 2024.06 **Visiting PhD Student** in Quantum Nano-Scale Magnetism Research Team.
RIKEN, Wako, Japan
- Advisor: Prof. YoshiChika OTANI
- 2018.09 – 2022.06 **Bachelor of Engineering degree** in Electronic Engineering.
University of Electronic Science and Technology of China, Chengdu, China
- Weighted Average: 90.15/100 GPA: 3.96/4.0
 - Core Courses:
Math Linear Algebra & Space Analytic Geometry: 93 Calculus: 92
Physics Semiconductor Physics A: 94 Microelectronic Device: 93
Signals and Systems (B): 95
Experiments EDA-Experiments: 98 Digital System Experiment I & II: 97&97

Professional Experience

- 2022.09 – Present **Postgraduate Research**
The Hong Kong University of Science and Technology, Hong Kong
Spintronic Quantum Material Laboratory
- Advisor: Prof. Qiming SHAO
 - Research topics: Quantum spintronic device; Photon-magnon, phonon-magnon and magnon-magnon hybrid system.
 - Leading Projects:
Strong magnon-magnon coupling in magnetic insulator bilayers [P2][P10]
Spin wave multiplier at zero magnetic field using quantum spin sensors [P1][P8]
Accurate model for coherent coupling between a magnon and a superconducting qubit [P7]
- 2023.11– 2024.06 **Visiting Ph.D. Student Research**
RIKEN, Wako, Japan
Quantum Nano-Scale Magnetism Research Team
- Advisor: Prof. YoshiChika OTANI & Prof. Qiming SHAO
 - Research topics: Hybrid magnon-phonon system
 - Leading Projects:
Hybrid magnon-phonon crystals [P3]

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2019.11– 2022.06 **Undergraduate Research**

University of Electronic Science and Technology of China, China

Integrated Microsystems Laboratory

- Advisors: Prof. Jingfu BAO & Prof. Kenya HASHIMOTO
- Research topics: Acoustic wave devices design; Phononic crystal
- Leading Projects:
Phononic crystal design to optimize the Q value of the acoustic wave resonator [P6][P7][P11][P12][P15]
Acoustic wave devices design, including SAW, BAW and FBAR [P13][P14][P16]

Honors and Awards

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| ● Outstanding Graduate of Sichuan Province (Top 1%), Si Chuan Province | 2021 |
| ● Outstanding Graduate of UESTC | 2021 |
| ● UESTC Outstanding Students (Top 10/4988 – 0.2%), UESTC | 2021 |
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| ● IEEE MSSS Distinguished Poster Award | 2025 |
| ● RedBird PhD Scholarship, HKUST | 2024 |
| ● Research Postgraduate Scholarship, HKUST | 2023 |
| ● RedBird PhD Scholarship, HKUST | 2023 |
| ● Student Presenter Travel Grant, IEDM | 2023 |
| ● UGC Research Travel Grant, HKUST | 2023 |
| ● Research Postgraduate Scholarship, HKUST | 2022 |
| ● UGC Research Travel Grant, HKUST | 2022 |
| ● RedBird PhD Scholarship, HKUST | 2022 |
| ● Gratitude to Modern Chinese Scientists Scholarship, Si Chuan Province | 2022 |
| ● National Inspiration Scholarship (top 1%), China | 2021 |
| ● Integrated Circuit Scholarship, China Electronics Society | 2021 |
| ● Merit Student Scholarship - First Class, UESTC | 2021 |
| ● Merit Student Scholarship - First Class, UESTC | 2020 |
| ● National Scholarship (Top 1%), China | 2019 |
| ● Merit Student Scholarship - First Class, UESTC | 2019 |
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| ● Gold Award of China College Students' "Internet+" Innov. & Entr. Comp. , China | 2022 |
| ● Global First Prize of Zhongguancun Talent Maker Comp. , Bei Jing | 2021 |
| ● First Prize of The Challenge Cup, Si Chuan Province | 2021 |

Professional Activities

Conference

- [C1] **“On-chip zero-field spin wave frequency multiplier and its application on qubit quantum control,”** 2025 IEEE Magnetic Society Summer School (IEEE MSSS), Poster, SanDigo, USA, June 2025 (*Poster Presentation*)
- [C2] **“Hybrid magnon phonon moiré lattices,”** 2025 Spin Summit, Poster, Gui Yang, China, August, 2025 (*Poster Presentation*)
- [C3] **“On-chip photon-mediated magnon-superconducting qubit system and its quantum application,”** 9th IEEE Electron Devices Technology and Manufacturing (EDTM), Oral- speaker, Hong Kong SAR, March 2025. (*Oral Presentation*)

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- [C4] **"On-chip zero-field spin wave frequency multiplier and its application on qubit quantum control,"** 2023 International Electron Devices Meeting (IEDM), Oral- speaker, San Francisco, CA, USA, December 2023. (*Oral Presentation*)
- [C5] **"Numerical and Micromagnetic Investigation of Strong Magnon-Magnon Coupling in Magnetic Insulator Bilayers,"** IEEE INTERMAG, Oral- speaker, Sendai, Japan, May 2023. (*Oral Presentation*)
- [C6] **"Acoustic Reflectors for Anchor Loss Reduction of Thin Film Piezoelectric on Substrate Resonators,"** IEEE MTT-S International Wireless Symposium, Oral- speaker, Nanjing, China May 2021. (*Oral Presentation*)
- [C7] **"Fish Scale-shaped Acoustic Reflector Array for Quality Factor Enhancement of AlN-on-Silicon MEMS Resonator,"** IEEE International Ultrasonics Symposium, Poster- presenter, Glasgow, England, October 2019. (*Poster Presentation*)

Seminar

- [S1]. **"Dissipative coupling in hybrid system based on magnonics,"** SQML Research seminar, Hong Kong, August 2024. (*Oral talk*)
- [S2]. **"Simulation of the phononic crystal (metamaterials) in COMSOL,"** SQML Research seminar, Hong Kong, August 2023. (*Oral talk*)
- [S3]. **"Strong Magnon-Magnon Coupling at the Interface of the Ferromagnetic Insulator Heterostructure,"** SQML Research seminar, Hong Kong, February 2022. (*Oral talk*)

Publications

Journal Papers

- [P1] Wu J*, Liu J*, Ren Z*, Leung M. Y, Leung W. K, Ho K. O, Shao Q, Yang S. **"Wideband coherent microwave conversion via magnon nonlinearity in a hybrid quantum system,"** npj Spintronics, 2(1), 30 (2024).
DOI: [10.1038/s44306-024-00035-2](https://doi.org/10.1038/s44306-024-00035-2)
- [P2] Liu J, Yuzan X, Jingming L, Xuezhao W, Chen L, Shun K C, Zheyu R, Ruizi L, Andrew C, Zehan C, Yifan L, Ferris P N, Xixiang Z, Dennis C W L, Zhang W, Shao Q. **"Strong magnon-magnon coupling and low dissipation rate in an all-magnetic-insulator heterostructure."** Physical Review Applied 22.3 (2024): 034017.
DOI: [10.1103/PhysRevApplied.22.034017](https://doi.org/10.1103/PhysRevApplied.22.034017)
- [P3] Liao L, Liu J, Puebla J, Shao Q, Otani Y. **"Hybrid magnon-phonon crystals,"** npj Spintronics, 2(1), 47 (2024).
DOI: [10.1038/s44306-024-00052-1](https://doi.org/10.1038/s44306-024-00052-1)
- [P4] Cheung S K, Xiao Z, Liu J, Ren Z, & Shao Q. **"Tunable intermediate states for neuromorphic computing with spintronic devices,"** Journal of Applied Physics, 136(4) (2024).
DOI: [10.1063/5.0187647](https://doi.org/10.1063/5.0187647)
- [P5] Ren Z, Liu R, Cheung S K, Qian K, Wu X, Xiao Z, Tong Z, Liu J, Shao Q. **"Strongly Temperature-Dependent Spin-Orbit Torque in a Sputtered Topological Semimetal,"** Journal of Applied Physics 135.14 (2024).
DOI: [10.1063/5.0157352](https://doi.org/10.1063/5.0157352)
- [P6] Liu J, Workie T B, Wu T, Wu Z, Gong K, Bao J, Hashimoto K. **"Q-factor Enhancement of Thin-Film Piezoelectric-on Silicon Mems Resonator by Phononic Crystal-Reflector Composite Structure"** [J]. Micromachines, 2020, 11(12): 1130.
DOI: [10.3390/mi11121130](https://doi.org/10.3390/mi11121130)

Conference Papers and Abstracts

- [P7] Liu J, Nugraha F. P, Shao Q. **"On-chip photon-mediated magnon-superconducting qubit**

- system and its quantum application,” 9th IEEE Electron Devices Technology and Manufacturing (EDTM), 2025 March 09 – March 12. (Oral, accepted)
- [P8] Liu J., Wu J, Ren Z, Yang S, Shao Q. **“On-chip zero-field spin wave frequency multiplier and its application on qubit quantum control,”** 2023 International Electron Devices Meeting (IEDM), San Francisco, CA, USA, 2023 December 09 – December 13. (Oral)
DOI: [10.1109/IEDM45741.2023.10413860](https://doi.org/10.1109/IEDM45741.2023.10413860)
- [P9] Liu J., Ohara Y, Xu Z, Liu R, Li Z, Wang X, Tsuchiya T, Wong M. **“Micro-Fabricated Bi-stable Mechanical Switch Actuated by Single Thermal Actuator,”** Transducers, Kyoto, Japan, 2023 June 7 – June 12. (Poster)
- [P10] Liu J., Shao Q. **“Numerical and Micromagnetic Investigation of Strong Magnon-Magnon Coupling in Magnetic Insulator Bilayers,”** IEEE INTERMAG, Sendai, Japan, 2023 May 15 – May 19. (Oral)
DOI: [10.1109/INTERMAGShortPapers58606.2023.10228275](https://doi.org/10.1109/INTERMAGShortPapers58606.2023.10228275)
- [P11] Ren Z, Liu R, Cheung S K, Qian K, Wu X, Xiao Z, Tong Z, Liu J., Shao Q. **“Strongly Temperature-Dependent Spin-Orbit Torque in Sputtered Wtex,”** APS March Meeting, Las Vegas, American, 2023 March 22 – June 25. (Oral)
- [P12] Liu J., Workie T B, Wu Z, Tang P, Bao J, Hashimoto K. **“Acoustic Reflectors for Anchor Loss Reduction of Thin Film Piezoelectric on Substrate Resonators,”** IEEE MTT-S International Wireless Symposium, Nanjing, China, 2021 May 23 – May 26. (Oral)
DOI: [10.1109/IWS52775.2021.9499583](https://doi.org/10.1109/IWS52775.2021.9499583)
- [P13] Workie T B, Liu J., Wu Z, Tang P, Bao J, Hashimoto K. **“Swastika Hole shaped Phononic Crystal for Quality Enhancement of Contour Mode Resonators,”** IEEE MTT-S International Wireless Symposium, Nanjing, China, 2021 May 23 – May 26. (Oral)
DOI: [10.1109/IWS52775.2021.9499380](https://doi.org/10.1109/IWS52775.2021.9499380)
- [P14] Gong K, Wu Z, Wong Y, Liu J., Liang Q, Li Y, Bao J, Hashimoto, K. **“Application of Phononic Crystal Structure for Side Leakage Suppression in A1-Mode Lamb Wave Resonators,”** IEEE International Ultrasonics Symposium, Xi’an, China, 2021 September 11 – September 16. (Oral)
DOI: [10.1109/IUS52206.2021.9593367](https://doi.org/10.1109/IUS52206.2021.9593367)
- [P15] Zhou X, Liu J., Bao F, Song Y, Bao J, Zhang X. **“Fish Scale-shaped Acoustic Reflector Array for Quality Factor Enhancement of AlN-on-Silicon MEMS Resonator,”** IEEE International Ultrasonics Symposium, Glasgow, England, 2019 October 7 – October 11. (Poster)
DOI: [10.1109/ULTSYM.2019.8926038](https://doi.org/10.1109/ULTSYM.2019.8926038)
- [P16] Wu X, Bao F, Zhou X, Wu Q, Liu J., Bao J. **“Spider Web-Shaped Phononic Crystals for Quality Factor Improvement of Piezoelectric-on-Silicon MEMS Resonators,”** IEEE International Ultrasonics Symposium, Glasgow, England, 2019 October 7 – October 11. (Poster)
DOI: [10.1109/ULTSYM.2019.8926212](https://doi.org/10.1109/ULTSYM.2019.8926212)

Patents

- [P17] Liu J., Hou S. **“Cellular Topology-Based Phononic Crystal Plate Design Method and Microelectromechanical Resonator,”** Chinese Invention Patent ZL 2021 1 0694788.0, CN 113507276 B, 2024.
- [P18] Shao Q., Yang S., Liu J., Wu J., Ren Z. *Spin-Wave Frequency Conversion Device and Application System.* Chinese Invention Patent Application, filed 2025, under examination. Assignee: The Hong Kong University of Science and Technology. (*first author in students*)

#UPDATE on 202602

